2. J. Phillips, *IT Project Management: On Track from Start to Finish*, 3rd ed. McGraw-Hill, Inc., 2010

SWE 4805 **Software Verification and Validation** **Credit: 3.00**

Introduction: Terminology, Evolving Nature of Area; V&V Limitations: Theoretical Foundations, Impracticality of Testing All Data, Impracticality of Testing All Paths, No Absolute Proof of Correctness, The Role of V&V in Software Evolution, V&V Objectives, Software V&V Approaches and their Applicability, Software V&V Planning, Organizational Responsibilities, Integrating V&V Approaches.

Recommended Texts:

1. M. S. Fisher, *Software Verification and Validation: An Engineering and Scientific Approach*, 1st ed. Springer Science & Business Media, 2007. DOI: 10.1007/978-0-387-47939-2 [Online]. Available: https://link.springer.com/book/10.1007/978-0-387-47939-2

CSE 4810 **Algorithm Engineering Lab** **Credit: 0.75**

Sessional works based on CSE 4809.

SWE 4802 **Software Maintenance Lab** **Credit: 0.75**

Sessional works based on SWE 4801.

SWE 4806 **Software Verification and Validation Lab** **Credit: 0.75**

Sessional works based on SWE 4805.

Elective 8-I (Selective)

SWE 4831 **OS Optimization and Real Time OS** **Credit: 3.00**

OS Concepts, Kernel, Micro Kernel vs Monolithic Kernel, Difference between OS and Real-Time(RT) OS, Relations and Differences of Embedded Systems with RT Systems, Introduction to concepts, techniques, and standards related to design of RT systems, Motivation, Specification of RT systems, Verification of RT systems, Optimizations in the Kernel, RT kernel architectures, Performance analysis of particular types of RT kernels, POSIX 1003.1b interface for RT operating systems (RTOS), Case Studies, RT task scheduling algorithms.

Recommended Texts:

1. D. Abbott, *Linux for Embedded and Real-time Applications*, 4th ed. Newnes, 2017

2. A. M. Cheng, *Real-Time Systems: Scheduling, Analysis, and Verification*, 1st ed. Wiley-Interscience, 2002

3. P. A. Laplante and S. J. Ovaska, *Real-Time Systems Design and Analysis: Tools for the Practitioner*, 4th ed. Wiley-IEEE Press, 2011

SWE 4833 **UI/UX Design** **Credit: 3.00**

Understanding User Experience (UX) & User Interface (UI) Design, Principles of good, Hicks law, Fitts's Law, General UI design workflow, and iterative design; Foundation of UX: Information Architecture, Elements of information architecture, Iconography, mapping user interaction, navigation structure; User interface design guidelines: Golden rules of design, principles of consistency and standards, visibility of the system status, error and slips control, recognition vs. recall, Aesthetic and Minimalist Design, Informative Feedback, Reduce Short-Term Memory Load, The Psychological Basis for UI Design Rules; Usability considerations: Simplicity in design, consistency in design, Don Norman design principles, User attention, chunking of information, understanding usability goals, measuring usability goals; HCI design models: Cognitive models, Workload models, Human information processing models, Distributed cognition models, Human Task Performance Measures; User experience modeling: Use case scenarios, writing scenarios, storytelling, building personas, mental model diagrams; UX and UI design: User-Centered Design (UCD), The Mobile, Web (And Desktop) Convergence, Responsive Design; Visual Elements: Understanding color psychology in design, color models, screen planning, sketching, wire-framing, prototyping, Icon and branding; Usability testing and evaluation: Usability evaluation, importance of user testing, Usability inspection methods, evaluating user interfaces.

Recommended Texts:

1. J. Johnson, *Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines*, 3rd ed. Morgan Kaufmann, 2020

2. J. Tidwell et al., *Designing Interfaces: Patterns for Effective Interaction Design*, 3rd ed. O'Reilly Media, 2020